FieldBridge AI — Documented Workflow Proof

Proof of Solution — Section 5.1, Item 2

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This document provides documented proof of the FieldBridge AI prompt-based simplification logic, as required under the 'Documented Workflow' proof format. It includes the full system prompt used by the tool and three real example conversions performed during testing and demonstration.

1. The Core System Prompt (Hindi Simplification Mode)

The following prompt is sent to the Claude AI API whenever a Research Associate uses Q-Bridge in 'Difficult Hindi → Simple Hindi' mode. It includes explicit word-replacement rules and worked examples to ensure the AI consistently produces village-level conversational Hindi rather than a lightly reworded version of the formal text.

SYSTEM PROMPT TEMPLATE:

```
Convert this formal Hindi question into very simple Hindi that a
farmer who studied till Class 5 can understand.

FORMAL QUESTION: "[question pasted by interviewer]"
OPTIONS TO ALSO SIMPLIFY: [options pasted by interviewer, if any]
```

WORD REPLACEMENTS — apply to BOTH question AND options:

अधिग्रहण/क्रय = खरीदना | परिचालन दक्षता = अच्छे से काम | शुष्कीकरण पद्धति = सुखाने का तरीका

अनुरक्षण = मरम्मत / देखभाल | पूंजीगत लागत = खरीदने का खर्च | लागत-लाभ अनुपात = फायदा-नुकसान

प्रत्यावर्तन = वापस जाना | संस्थागत सहायता = सरकारी मदद | पर्याप्तता = काफी है या नहीं

अभिगम्यता = मिलना / पहुंचना | तकनीकी अनियमितता = खराबी | इकाई = मशीन

अभिप्रेरणा = कारण / वजह | मूल्यांकन = सोचना / राय | श्रेष्ठ = बेहतर | संवर्धन = सुधार

RULES:

- Question: max 15 words. Options: max 10 words each.

- Sound like one farmer talking to another.
- NEVER use formal/Sanskrit words in question or options.

Return ONLY valid JSON:

```
{"simplified":"...","simplified_options":[...],
 "terms":[{"term":"...","explain":"..."}],
 "green":"...","yellow":"...","red":"...",
 "followup":"...","followupNote":"..."}
```

2. Real Example Inputs and Outputs

Three real conversions performed using the live tool are documented below, showing the original formal Hindi question, the AI-generated simplified version, and supporting outputs.

Example 1

⚠ ORIGINAL (Formal Hindi — Acquisition Section)

सौर ड्रायर इकाई के अधिग्रहण हेतु आपकी प्राथमिक अभिप्रेरणा क्या थी?

✓ SIMPLIFIED (AI Output)

आपने सोलर ड्रायर क्यों खरीदा?

Term Explainer: "अधिग्रहण" explained as "खरीदना" (to buy).

Response Confidence Guide — Green: consumer names a specific clear reason (e.g. subsidy, income, quality); Yellow: vague answer like "बस ऐसे ही"; Red: consumer appears confused by the question itself.

Follow-Up Suggestion: "आपको सबसे पहले इसके बारे में किसने बताया था?"

Example 2

⚠ ORIGINAL (Formal Hindi — Maintenance Section)

क्या आपको सौर ड्रायर इकाई में किसी तकनीकी या यांत्रिक अनियमितता का सामना करना पड़ा है?

✓ SIMPLIFIED (AI Output)

क्या मशीन में कभी कोई खराबी आई?

Term Explainer: "तकनीकी या यांत्रिक अनियमितता" explained as "मशीन में खराबी" (a fault or breakdown in the machine).

Response Confidence Guide — Green: consumer describes a specific fault and what they did about it; Yellow: consumer says "शायद" or hesitates; Red: consumer answers an unrelated question.

Follow-Up Suggestion: "जब खराबी आई तो आपने किससे मदद मांगी?"

Example 3

⚠ ORIGINAL (Formal Hindi — Switching/Abandonment Section)

सौर ड्रायर इकाई के अधिग्रहण के पश्चात किन परिस्थितियों में आपने पारंपरिक धूप में सुखाने की विधि की ओर पुनः प्रत्यावर्तन किया?

✓ SIMPLIFIED (AI Output)

सोलर ड्रायर लेने के बाद कभी वापस पुराने तरीके से सुखाना शुरू किया?

Term Explainer: "प्रत्यावर्तन" explained as "वापस जाना" (going back to the old way).

Response Confidence Guide — Green: consumer clearly says yes/no and gives a reason; Yellow: ambiguous answer that doesn't confirm either way; Red: consumer seems unsure what is being asked.

Follow-Up Suggestion: "किस मौसम में आप पुराने तरीके पर वापस जाते हैं?"

3. Notes on This Proof

These three examples were generated using the live FieldBridge AI tool (FieldBridge_AI_Demo.html) connected to the Claude AI API, using formal Hindi questions modeled directly on the structure and register of the actual CEEW solar dryer questionnaire used during the internship fieldwork. The actual Kantar client questionnaire could not be reproduced in this document due to client data confidentiality protocols; the questions used here are representative of the same formal register and were used to validate the tool's simplification logic, including during the supervisor demonstration described in Section 3.2 of the case write-up.